

Learning from Rankings ($m \leq 3$)
Unidentifiable

Distributions of Rankings ($m = 3$)

$$P(A \succ C \succ D) = P(A \succ D \succ C) = 1/8$$

$$P(A \succ B \succ C) = P(B \succ A \succ C) = 1/8$$

$\vdots$

Distributions of Rankings ($m = 2$)

$$P(A \succ C) = P(A \succ D) = 1/6$$

$$P(B \succ C) = P(B \succ D) = 1/6$$

$\vdots$

Underlying Clusters

True Clusters

50%: $A \succ B \succ C \succ D$

50%: $B \succ A \succ D \succ C$

Alternative Clusters

50%: $A \succ B \succ D \succ C$

50%: $B \succ A \succ C \succ D$

Learning from Rankings ($m = 4$)
Identifiable

Distributions of Rankings ($m = 4$)

$$P(A \succ B \succ C \succ D) = 50\%$$

$$P(B \succ A \succ D \succ C) = 50\%$$

Distributions of Rankings ($m = 4$)

$$P(A \succ B \succ D \succ C) = 50\%$$

$$P(B \succ A \succ C \succ D) = 50\%$$